

Literature Databases, Searching and Reference Management

150 minutes · 6 self-check questions

Jesse Anak · Institute of Computational Biology and Bioinformatics

The databases that matter, how to build a search that is both sensitive and precise, how to document it for PRISMA, and how to manage references so the bibliography is never done by hand.

Learning objectives

- Select appropriate databases for a given topic and justify the choice
 - Construct a Boolean search using controlled vocabulary and field tags
 - Use MeSH terms, truncation, phrase searching and proximity operators
 - Balance sensitivity against precision and explain the trade-off
 - Document a search so that another researcher can reproduce it exactly
 - Set up Zotero or Mendeley and generate a formatted bibliography automatically
-

The major databases

- PubMed / MEDLINE — biomedical and life sciences; free; indexed with MeSH controlled vocabulary
- Scopus and Web of Science — large multidisciplinary citation databases; subscription; best for citation tracking and bibliometrics
- Cochrane Library — controlled trials and systematic reviews; essential for effectiveness questions
- CINAHL — nursing and allied health; EMBASE — stronger European and pharmacological coverage than PubMed
- PsycINFO — psychology and behavioural sciences; ERIC — education
- African Journals Online (AJOL) — the largest collection of African-published scholarship, poorly covered by the big indexes
- Google Scholar — very broad and useful for finding grey literature and chasing citations, but the algorithm is opaque and results are not reproducible, so it should supplement rather than replace a structured search

IN PRACTICE

For work on African populations, searching only PubMed and Scopus will systematically miss regionally published evidence. AJOL, national journal portals and institutional repositories are not optional extras here.

Building the search

- Break the question into concepts — usually the P, I/E and O of PICO; do not search the whole question as a sentence
- For each concept, gather synonyms, spelling variants, abbreviations and older terminology
- Combine synonyms within a concept using OR; combine concepts using AND; use NOT sparingly, as it discards relevant records
- Truncation: malari* retrieves malaria, malarial, malariae
- Phrase searching: "mobile money" keeps the words adjacent
- Field tags restrict where a term is searched, e.g. [tiab] for title and abstract in PubMed
- Proximity operators (NEAR, W/n, ADJn) find terms near each other and vary by database

Controlled vocabulary

- MeSH is the National Library of Medicine's controlled vocabulary; each PubMed record is tagged by indexers
- Searching a MeSH term retrieves records regardless of the authors' wording
- Explosion automatically includes narrower terms beneath a heading in the tree
- Major topic restricts to records where the concept is a main focus
- MeSH indexing lags publication, so very recent papers are found only by free-text terms
- The reliable strategy combines both: MeSH OR free-text, for each concept
- Emtree in EMBASE and CINAHL Headings serve the same purpose in their databases

SENSITIVITY VERSUS PRECISION

Goal	Strategy	Cost
High sensitivity (systematic review)	Many synonyms, truncation, explode MeSH, no limits	Thousands of records to screen
High precision (quick scoping)	Few terms, major topic, title field, recent years	Relevant studies will be missed
Balanced (thesis chapter)	MeSH OR free-text per concept, sensible date limit	Manageable; state the limits used

Documenting the search

- Record for each database: its name, the platform, the date searched, the full strategy line by line, and the number of records returned
- The date matters because databases change daily and the search must be reproducible
- Report screening with a PRISMA flow diagram: identified, duplicates removed, screened, excluded with reasons, included
- Keep the exclusion reasons — reviewers ask for them
- Save the strategy in the database's own account feature and export it, rather than retyping later
- Set up email alerts so new hits arrive during your fieldwork

Reference management

- Zotero — free and open source, excellent browser capture, best default for most researchers
- Mendeley — free, good PDF annotation, owned by Elsevier
- EndNote — paid, common in institutions with a site licence, strong for very large libraries
- All three insert citations into Word or LibreOffice and reformat the entire bibliography when you change style
- Deduplicate on import, and fix metadata at import time — a wrong year propagates into every citation
- Use collections per chapter and tags per theme; attach the PDF and your notes to the record
- Back up the library, and sync it — a lost reference library near submission is a genuine disaster
- Never type a bibliography by hand: it wastes days and introduces errors that examiners do notice

IN PRACTICE

Set your citation style once at the beginning. Switching from APA to Vancouver later is a single dropdown in the reference manager, and a week of manual work without one.

Worked examples

Building a PubMed search line by line

The question is whether community health worker home visits reduce childhood malaria in rural Ghana. Typing the whole sentence into PubMed returns almost nothing useful.

1. Split into concepts: (1) community health workers, (2) home visits, (3) malaria in children.
2. Concept 1: "Community Health Workers"[Mesh] OR community health worker*[tiab] OR CHW[tiab] OR "village health volunteer*" [tiab]
3. Concept 2: "House Calls"[Mesh] OR home visit*[tiab] OR domiciliary[tiab]
4. Concept 3: "Malaria"[Mesh] AND ("Child"[Mesh] OR child*[tiab] OR under-five*[tiab])
5. Combine: #1 AND #2 AND #3. Record the date, the number of hits and the exact strings.
6. Then rerun the concepts in AJOL and Google Scholar — the Ghanaian and regional literature is thin in PubMed and you will otherwise miss it.

THE POINT

Search concepts, never sentences. And a search you cannot reproduce next month, because you did not record it, is not a method — it is a memory.

What truncation actually costs you

A researcher searches nurs expecting to capture nurse, nurses and nursing. The result set balloons and is full of irrelevant records.*

1. nurs* also matches nursery, nurseries and nursed, none of which are wanted.
2. Truncating too early is the usual cause: nurs* cuts before the stem is distinctive.
3. nurse* OR nursing captures what is needed without the nursery records.
4. Check the effect: run both and compare the hit counts and a sample of titles.
5. Where a database offers it, a field tag narrows further — nurse*[tiab] rather than a free-text search of the whole record.

THE POINT

Truncate at the point where the stem is still unambiguous, then look at what you actually retrieved. Wildcards are not free.

Key terms

MeSH

Medical Subject Headings — the controlled vocabulary used to index MEDLINE and PubMed records.

Boolean operators

AND, OR and NOT, used to combine search concepts.

Truncation

A wildcard symbol (usually *) retrieving all word endings from a common stem.

Sensitivity

The proportion of all relevant records that a search retrieves.

Precision

The proportion of retrieved records that are actually relevant.

PRISMA flow diagram

The standard figure reporting how records moved from identification through screening to inclusion.

Grey literature

Reports, theses and working papers published outside commercial channels and often outside major indexes.

Activity

Build a full PubMed strategy for your own question. Produce at least two concepts, each with MeSH and free-text synonyms combined by OR, then joined by AND. Record the date and the number of hits. Import the first twenty results into Zotero and generate a bibliography in APA, then switch it to Vancouver.

Further reading

- **PubMed User Guide — search field tags and MeSH** — Free, authoritative and short.
- **Bramer et al. (2018) — "A systematic approach to searching"** — A practical, reproducible method for building strategies.
- **Cochrane Handbook, Chapter 4 — Searching for studies** — The reference standard for exhaustive searching.
- **Zotero documentation — Quick Start Guide** — Enough to be productive in under an hour.

Self-check questions

1. To find records using either term for the same concept, you should combine "adolescent" and "teenager" with:
 - a. AND
 - b. OR

- c. NOT
 - d. NEAR
2. The main advantage of searching with MeSH terms rather than free text alone is that:
- a. MeSH searches are faster
 - b. Records are retrieved regardless of the wording the authors chose
 - c. MeSH always returns fewer results
 - d. MeSH includes the most recently published papers first
3. A researcher studying maternal health services in Ghana searches only PubMed and Scopus. The most likely consequence is that they:
- a. Retrieve too many irrelevant records
 - b. Miss regionally published African evidence indexed in AJOL and national repositories
 - c. Violate PRISMA reporting rules
 - d. Introduce recall bias
4. A systematic review search should favour:
- a. Precision, to keep screening manageable
 - b. Sensitivity, accepting a larger screening burden
 - c. Google Scholar alone, for breadth
 - d. Only papers published in the last five years
5. Why must the date of each database search be recorded?
- a. To calculate the response rate
 - b. Because databases are updated continuously, so the search is only reproducible as at a given date
 - c. It is required for ethical approval
 - d. To determine the citation style
6. The strongest practical reason to use a reference manager is that it:
- a. Improves the quality of your writing
 - b. Reformats every citation and the whole bibliography when the required style changes
 - c. Guarantees you avoid plagiarism
 - d. Finds relevant papers for you

Answers

1. **b. OR** broadens by retrieving records containing either synonym, so it combines terms within a concept. **AND** narrows and is used between different concepts.

2. **b.** Indexers assign MeSH consistently, so a MeSH search captures records however the authors phrased it. Its weakness is the opposite: very recent records are not yet indexed, which is why MeSH and free text are combined with OR.
3. **b.** Coverage of African-published journals in the major international indexes is incomplete. AJOL, institutional repositories and national portals are needed to avoid systematically missing local evidence.
4. **b.** Systematic reviews aim to find all relevant studies, so they prioritise sensitivity and accept screening thousands of titles. Missing eligible studies biases the conclusion; screening extra records only costs time.
5. **b.** New records are added daily. Without the search date, another researcher rerunning your strategy cannot explain why their result count differs, and reproducibility is lost.
6. **b.** Style switching is the decisive practical gain — a dropdown instead of days of manual reformatting. It does not write for you, and it does not by itself prevent plagiarism.

